

University of Prishtina “Hasan PRISHTINA”



Faculty of Electrical and Computer Engineering
Master in Computer and Software Engineering, 2025-2026
Course: Data Preparation and Visualization

Week VI – Anomaly Detection I

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Anomaly detection

Anomalies and outliers

**are essentially the same thing:
objects that are different from most other objects**

The techniques used for detection are the same.

Anomaly detection

- Historically, the field of statistics tried to find and remove outliers as a way to improve analyses.
- There are now many fields where the outliers / anomalies are the objects of greatest interest:
 - The rare events may be the ones with the greatest impact, and often in a negative way.

Causes of anomalies

- Data from different class of object or underlying mechanism:
 - disease vs. non-disease;
 - fraud vs. not fraud.

- Natural variation:
 - tails on a Gaussian distribution.

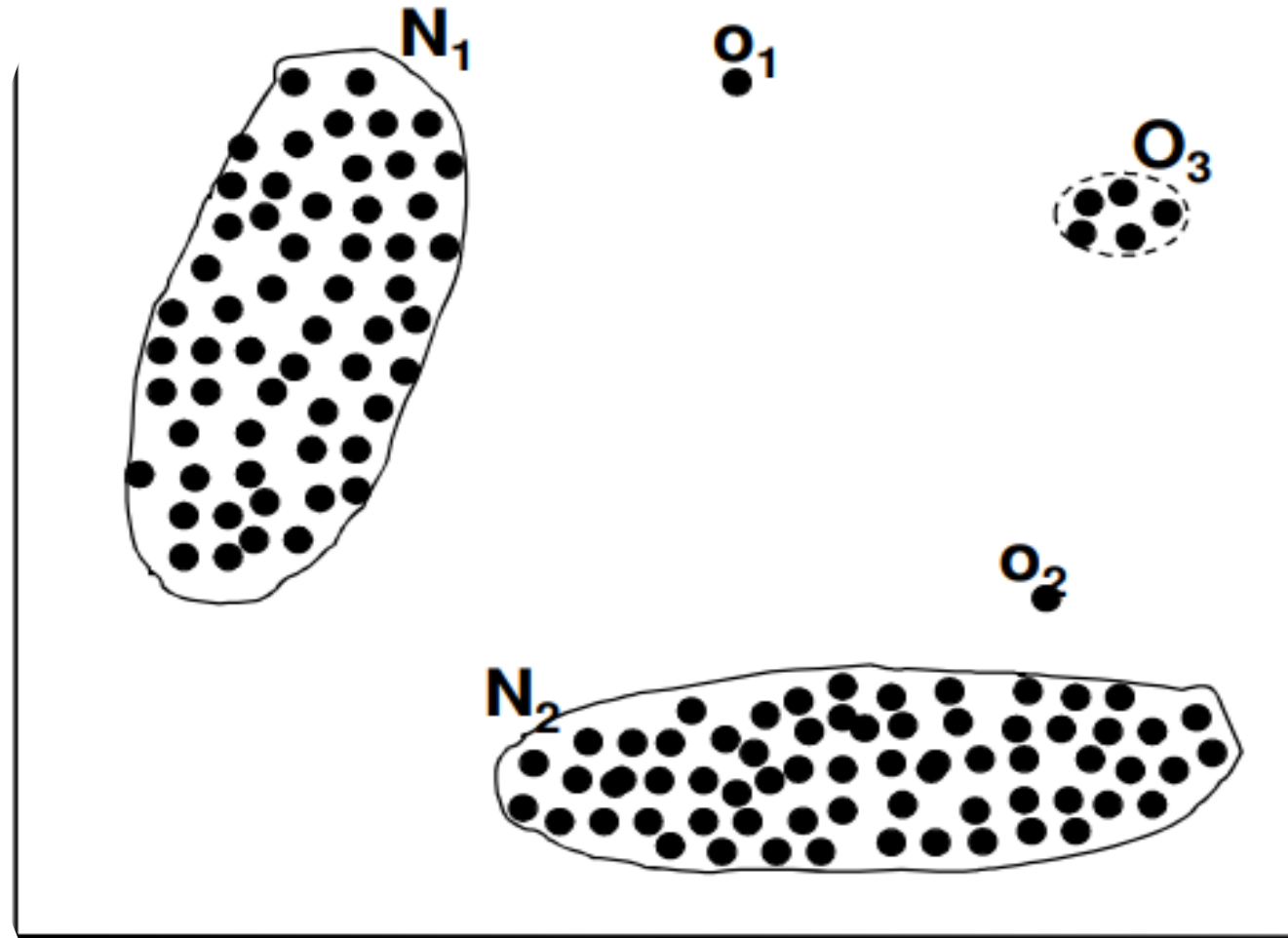
- Data measurement and collection errors.

Structure of anomalies

- Point anomalies;
- Contextual anomalies;
- Collective anomalies.

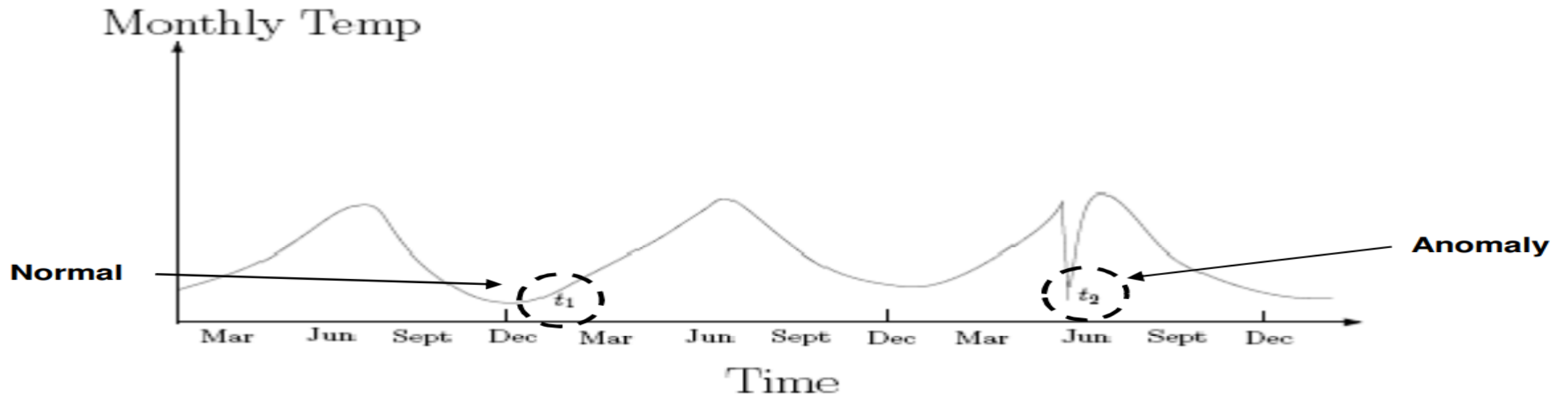
Point anomalies

- An individual data instance is anomalous with respect to the data.



Contextual anomalies

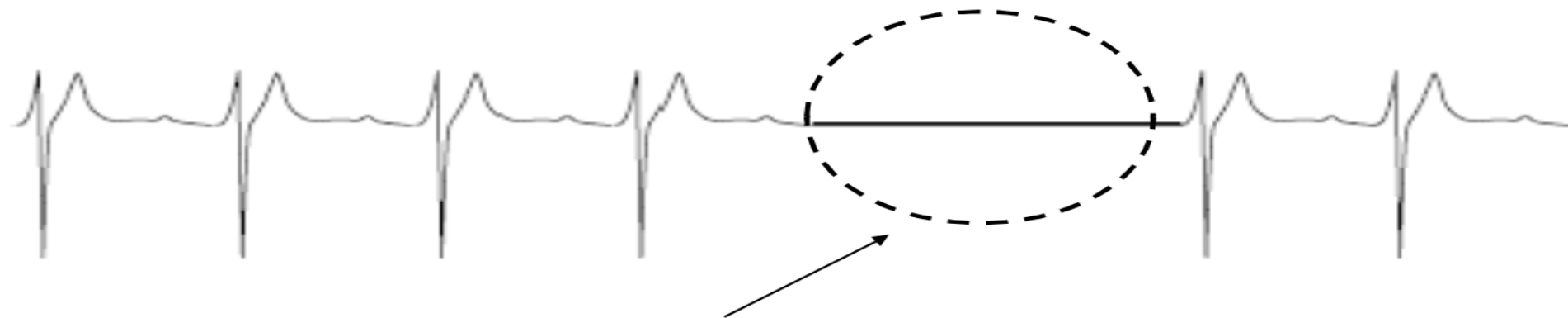
- An individual data instance is anomalous within a context;
- Requires a notion of context;
- Also referred to as conditional anomalies.*



* Song, et al, “Conditional Anomaly Detection”, IEEE Transactions on Data and Knowledge Engineering, 2006

Collective anomalies

- A collection of related data instances is anomalous;
- Requires a relationship among data instances:
 - Sequential data;
 - Spatial data;
 - Graph data.
- The individual instances within a collective anomaly are not anomalous by themselves.



anomalous subsequence

Applications of anomaly detection

- Network intrusion;
- Insurance/credit card fraud;
- Healthcare informatics / medical diagnostics;
- Industrial damage detection;
- Image processing/video surveillance;
- Novel topic detection in text mining;
- ...

Intrusion detection

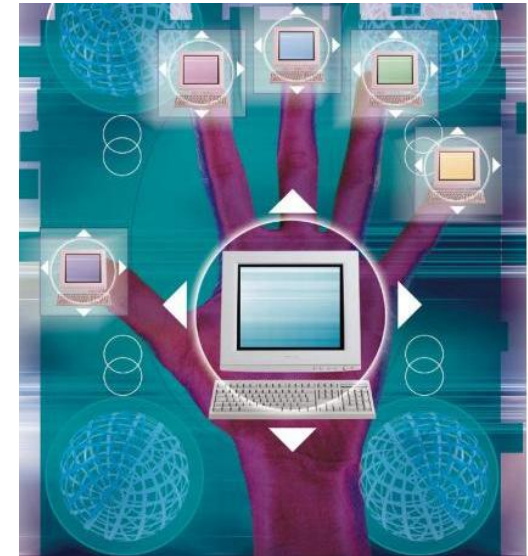
➤ Intrusion detection:

- Monitor events occurring in a computer system or network and analyze them for intrusions;
- Intrusions defined as attempts to bypass the security mechanisms of a computer or network.

➤ Challenges:

- Traditional intrusion detection systems are based on signatures of known attacks and cannot detect emerging cyber threats;
- Substantial latency in deployment of newly created signatures across the computer system.

➤ Anomaly detection can alleviate these limitations.



Fraud detection

- Detection of criminal activities occurring in commercial organizations;
- Malicious users might be:
 - Employees;
 - Actual customers;
 - Someone posing as a customer (identity theft).
- Types of fraud:
 - Credit card fraud;
 - Insurance claim fraud;
 - Mobile / cell phone fraud;
 - Insider trading.
- Challenges:
 - Fast and accurate real-time detection;
 - Misclassification cost is very high.



Healthcare informatics

➤ Detect anomalous patient records:

- Indicate disease outbreaks;
- Instrumentation errors, etc.

➤ Key challenges:

- Only normal labels available;
- Misclassification cost is very high;
- Data can be complex: *spatio-temporal*.



Industrial damage detection

➤ Detect faults and failures in complex industrial systems, structural damages, intrusions in electronic security systems, suspicious events in video surveillance, abnormal energy consumption, etc.

- Example: aircraft safety

- anomalous aircraft (engine) / fleet usage;
- anomalies in engine combustion data;
- total aircraft health and usage management.

➤ Key challenges:

- Data is extremely large, noisy, and unlabelled;
- Most of applications exhibit temporal behaviour;
- Detected anomalous events typically require immediate intervention.



Image processing

- Detecting outliers in an image monitored over time;
- Detecting anomalous regions within an image;
- Used in:
 - mammography image analysis;
 - video surveillance;
 - satellite image analysis.
- Key Challenges:
 - Detecting collective anomalies;
 - Datasets are very large.



Use of data labels in anomaly detection

- Supervised anomaly detection:
 - Labels available for both normal data and anomalies;
 - Similar to classification with high class imbalance.
- Semi-supervised anomaly detection:
 - Labels available only for normal data.
- Unsupervised anomaly detection:
 - No labels assumed;
 - Based on the assumption that anomalies are very rare compared to normal data.

Output of anomaly detection

➤ Label:

- Each test instance is given a normal or anomaly label;
- Typical output of classification-based approaches.

➤ Score:

- Each test instance is assigned an anomaly score:
 - allows outputs to be ranked;
 - requires an additional threshold parameter.

Variants of anomaly detection problem

- Given a dataset D , find all the data points $x \in D$ with anomaly scores greater than some threshold t ;
- Given a dataset D , find all the data points $x \in D$ having the top- n largest anomaly scores;
- Given a dataset D , containing mostly normal data points, and a test point x , compute the anomaly score of x with respect to D .

Unsupervised anomaly detection

- No labels available;
- Based on assumption that anomalies are very rare compared to “**normal**” data;
- General steps:
 - Build a profile of “normal” behaviour:
 - summary statistics for overall population;
 - model of multivariate data distribution.
 - Use the “normal” profile to detect anomalies:
 - anomalies are observations whose characteristics differ significantly from the normal profile.

Techniques for anomaly detection

- Statistical;
- Proximity-based;
- Density-based;
- Clustering-based.

Statistical outlier detection

Outliers are objects that are **fit** poorly by a statistical model.

- Estimate a parametric model describing the distribution of the data;
- Apply a statistical test that depends on:
 - Properties of test instance;
 - Parameters of model (e.g., mean, variance);
 - Confidence limit (related to number of expected outliers).

Questions?

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